MATHEMATICAL AND COMPUTATIONAL REPORT

Expected Differences and Ranges of Correlated IQ Scores

Two tests, five tests, and the limits of correlation

AI-GENERATED REPORT: READ THIS FIRST

ChatGPT produced the research synthesis, mathematical exposition, source code, figures, and document in response to Anders Hellström's request. The simulations are synthetic; no observed IQ-test dataset was analysed. The numerical answers depend on explicit assumptions and have not received independent human peer review. A complete AI-use disclosure appears in the final appendix.

At SD 15 and pairwise correlation 0.6, under joint normality:

Expected absolute difference between two tests	**10.70 points**
Expected highest-to-lowest range across five tests	**22.07 points**

Correlation alone does not determine either expectation.

Prepared at the request of **Anders Hellström**

Research and computation: **3 September 2026**
Authoring system: ChatGPT / Codex, OpenAI
Format: XeLaTeX, KOMA-Script `scrbook`; A4, narrow margins

Contents

1 Executive findings

The direct answer

If IQ scores have common standard deviation 15, are jointly normal, and every distinct pair has correlation 0.6, then

$$\boxed{\mathbb{E}|X_1 - X_2| = 10.7047446969 \text{ points}}, \qquad \boxed{\mathbb{E}(\max_{1\le j\le 5} X_j - \min_{1\le j\le 5} X_j) = 22.0656994474 \text{ points}}.$$

The expected signed difference is zero when the two means are equal.

These are exact expectations under a specified probability model. They are not empirical estimates obtained from a sample of people taking real IQ tests. The decimal values are numerical evaluations of proved formulas; reporting 10.70 and 22.07 points is normally sufficient.

The question continued from the preceding analysis is: what is the average difference between two IQ tests when their correlation is 0.6, and what is the average maximum-minus-minimum across five such tests? The essential task is to identify what follows from correlation, what needs a distributional assumption, and how simulation can check the implementation without replacing a proof.

1.1 The assumptions that make the question determinate

Component	Assumption used for the main answer
Population	A randomly selected person from one defined population takes the same fixed set of tests.
Score scale	Every test has the same mean μ and SD $\sigma = 15$; $\mu = 100$ when score levels are shown.
Dependence	The entire score vector is multivariate normal.
Correlation	Each of the ten distinct pairs among five tests has population Pearson correlation $\rho = 0.6$.
Reporting	Scores are continuous and unrounded; no floor, ceiling, occasion bias, or practice trend is added.
Averaging	Expectations are unconditional across people, not conditional on a previously observed score.

The mean-100, SD-15 convention is used for the WISC-V full-scale IQ and index scores, for example, but not its subtest scaled scores (NCS Pearson, 2020). The convention does not itself establish normality, equal intertest correlations, or applicability to a selected population. No test names or observed correlation matrix were supplied here: 0.6 is a stipulated parameter, not a literature estimate that this report validates.

1.2 Findings beyond the two averages

1. **Correlation fixes the variance of the two-test difference, but not its mean absolute value.** With the stated first two moments, $\mathbb{E}[(X_1 - X_2)^2] = 180$ square IQ points, even without normality.
2. **Individual variation is substantial.** In the Gaussian model, the median five-test range is 21.41 points; its central 95% interval is approximately 8.06–39.82 points. This is a distribution interval, not uncertainty about the mean 22.07.

3. **Normal marginals are insufficient.** Two alternative constructions retain all five normal marginal distributions and every pair correlation 0.6, yet produce different expectations. One is a smooth mixture with no point mass at identical scores.
4. **An average correlation of 0.6 is insufficient.** Even within the Gaussian family, a different correlation pattern with that average can give an expected five-test range of 13.82 points.
5. **A known first score changes the question.** If the first score is 145, the Gaussian model predicts a mean second score of 127 and a mean absolute gap of 18.70 points, rather than the unconditional 10.70.
6. **The code reproduces the numerical claims.** A fresh run of five million independent people per main model reproduces the preceding run exactly; the two requested Gaussian means are within one Monte Carlo standard error of their analytic values. A separate million-person run checks the smooth mixture.

1.3 How to read the report

Chapters 2–4 establish the model and both expectation formulas, including the closed form for five tests. Chapters 5–7 give individual distributions, counterexamples, conditioning, and sensitivity. Chapters 8–9 discuss psychometric interpretation and computational evidence. Chapter 10 records the research scope and remaining uncertainties. The bibliography contains 21 sources, followed by reproducibility details, complete program listings, and the full AI disclosure as the final appendix.

2 The model and what correlation establishes

2.1 Definitions and the unit of analysis

Let $\mathbf{X} = (X_1, \dots, X_k)^\top$ be the scores of one randomly selected person on k fixed tests. Define

$$D = X_1 - X_2, \qquad A = |D|, \qquad R_k = \max_j X_j - \min_j X_j.$$

Independent people provide independent replications of this vector. Scores within a person are dependent. The range is always nonnegative and is also $\max_{i<j} |X_i - X_j|$; it is not the average of the pairwise absolute gaps.

Pearson correlation is $\operatorname{Corr}(X_i, X_j) = \operatorname{Cov}(X_i, X_j)/(\sigma_i \sigma_j)$. It describes standardized covariance, not equality of scores. For instance, $Y = X + 20$ has correlation one with X but a constant 20-point difference. This simple algebra exemplifies the association/agreement distinction made by Bland and Altman (1986).

If both tests have mean μ, SD σ, and correlation ρ, then

$$\mathbb{E}D = 0, \tag{2.1}$$
$$\operatorname{Var}(D) = \operatorname{Var}(X_1) + \operatorname{Var}(X_2) - 2\operatorname{Cov}(X_1, X_2) = 2\sigma^2(1-\rho). \tag{2.2}$$

No normality is needed for these identities. At $\sigma = 15$ and $\rho = 0.6$, the difference variance is 180 and its SD is $\sqrt{180} = 13.416407865$ points. Consequently the root-mean-square difference is determined from these moments. The absolute first moment $\mathbb{E}|D|$ is not yet determined.

2.2 Joint normality and equicorrelation

The main model is

$$\mathbf{X} \sim N_k\big(\mu\mathbf{1}, \sigma^2[(1-\rho)I_k + \rho\mathbf{1}\mathbf{1}^\top]\big). \tag{2.3}$$

A vector is jointly normal when every real linear combination of its components is univariate normal, allowing degenerate normal variables at boundary cases. The linear-transformation property and the distinction from marginal normality are discussed in Pishro-Nik (2014) and Taboga (n.d.).

For $0 \le \rho \le 1$, take independent standard normals $Z_0, Z_1, \dots, Z_k$ and put

$$X_j = \mu + \sigma\sqrt{\rho}\, Z_0 + \sigma\sqrt{1-\rho}\, Z_j. \tag{2.4}$$

This vector is jointly normal. Each coordinate has variance $\sigma^2\rho + \sigma^2(1-\rho) = \sigma^2$, and two distinct coordinates share only the common term, so their covariance is $\sigma^2\rho$. Its characteristic function is

$$\mathbb{E}e^{i\mathbf{t}^\top\mathbf{X}} = \exp\big(i\mu\mathbf{t}^\top\mathbf{1} - \tfrac{1}{2}\mathbf{t}^\top\Sigma\mathbf{t}\big).$$

Thus the mean vector and covariance matrix determine the entire Gaussian law. The construction represents the assumed law, rather than an arbitrary example among many different Gaussian laws with that same covariance matrix.

The common component is a convenient mathematical representation of shared variation. The algebra does not establish that it is a person's true IQ, or that every test-specific component is measurement error.

2.3 Which equicorrelations are possible?

The equicorrelation matrix has eigenvalue $1 + (k-1)\rho$ in the direction $\mathbf{1}$ and eigenvalue $1 - \rho$ on its orthogonal complement. Both must be nonnegative, giving

$$-\frac{1}{k-1} \leq \rho \leq 1. \tag{2.5}$$

The interior is positive definite; the endpoints may be singular. The eigenvalue characterization is also stated by Archakov et al. (2024), in the public preliminary version consulted for this report.

The familiar construction in equation (2.4) uses $\sqrt{\rho}$, so it does not apply to negative correlations. A construction that covers the full admissible interval is

$$X_j = \mu + \sigma\sqrt{1-\rho}(Z_j - \bar{Z}) + \sigma\sqrt{\frac{1+(k-1)\rho}{k}}\,U, \tag{2.6}$$

where $U, Z_1, \dots, Z_k$ are independent standard normals and $\bar{Z} = k^{-1}\sum_j Z_j$. The centered vector has covariance $I_k - \mathbf{1}\mathbf{1}^\mathsf{T}/k$. Adding the last independent term gives exactly the covariance in equation (2.3). For five tests, the lower endpoint is $-1/4$. The main simulation script deliberately supports only $0 \leq \rho \leq 1$; the supplement checks the general covariance construction.

3 The absolute difference between two tests

Theorem 3.1 (Two-test absolute difference)**.** *For jointly normal scores with a common mean and SD σ, and correlation $\rho \in [-1, 1]$,*

$$\boxed{\mathbb{E}|X_1 - X_2| = 2\sigma\sqrt{\frac{1-\rho}{\pi}}.} \tag{3.1}$$

Proof. The linear contrast $D = X_1 - X_2$ is normal, with mean zero and variance from equation (2.2). Write $\tau = \sigma\sqrt{2(1-\rho)}$. If $\tau = 0$, the difference is zero almost surely and the formula follows. Otherwise symmetry gives

$$\begin{aligned}\mathbb{E}|D| &= \frac{2}{\tau\sqrt{2\pi}} \int_0^\infty t \exp\left(-\frac{t^2}{2\tau^2}\right) \mathrm{d}t \\ &= \frac{2}{\tau\sqrt{2\pi}}\tau^2 = \tau\sqrt{\frac{2}{\pi}}.\end{aligned}$$

The integral is τ^2 by the substitution $u = t^2/(2\tau^2)$. Substituting the definition of τ proves the result. $\square$

Numerically,

$$\mathbb{E}A = 30\sqrt{0.4/\pi} = 10.7047446969\ldots.$$

The absolute difference has a half-normal distribution, because it is the absolute value of a centered normal variable. From $A^2 = D^2$,

$$\mathrm{Var}(A) = \tau^2\left(1 - \frac{2}{\pi}\right), \qquad \mathrm{SD}(A) \approx 8.0875 \text{ points.} \tag{3.2}$$

This population SD describes the variability of individual absolute gaps. It should not be confused with the simulation mean's standard error, which is smaller by a factor $\sqrt{n}$ for n independent people.

3.1 Signed, absolute, and root-mean-square differences

Quantity	**IQ points**	**Meaning**
$\mathbb{E}D$	0	Positive and negative differences cancel.
$\mathbb{E}\lvert D\rvert$	10.7047	Average magnitude of the gap.
$\sqrt{\mathbb{E}D^2}$	13.4164	Square, average, and take the square root.
median($\lvert D\rvert$)	9.0492	Half of absolute gaps are below this value.

The RMS difference exceeds the mean absolute difference because larger gaps receive greater weight when squared. The signed mean being zero does not mean that two scores normally coincide.

3.2 Unequal means or standard deviations

The equal-scale assumption can be relaxed for a jointly normal pair. Let $\delta = \mu_1 - \mu_2$ and $\tau^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$. Then $D \sim N(\delta, \tau^2)$. For $\tau > 0$, splitting the expectation at zero gives

$$\mathbb{E}|D| = 2\tau\phi(\delta/\tau) + \delta\{2\Phi(\delta/\tau) - 1\}, \tag{3.3}$$

where ϕ and Φ are the standard normal density and CDF. To verify this, write $\mathbb{E}|D| = \mathbb{E}D - 2\mathbb{E}[D\mathbf{1}_{D<0}]$ and substitute $D = \delta + \tau Z$. The truncated moment is $\delta\Phi(-\delta/\tau) - \tau\phi(\delta/\tau)$. For $\tau = 0$, the answer is $|\delta|$.

Thus systematic score offsets and unequal scales require different numerical answers even when the correlation remains 0.6. Marginal standardization changes the question to differences in standardized units; it does not erase such differences on the original reported score scales.

4 The range across five tests: exact proof

4.1 Why the shared component cancels

Put $s = \sigma\sqrt{1-\rho}$, and define the range of independent standard normals as $W_k = \max_j Z_j - \min_j Z_j$. From equation (2.4),

$$R_k = sW_k. \tag{4.1}$$

The common mean and common normal component cancel for every realization, before any expectation is taken. Equation (2.6) gives the same identity throughout the admissible negative-correlation interval: the $-s\bar{Z}$ and U terms also cancel.

Let $M_k = \max_j Z_j$ and $m_k = \min_j Z_j$. Simultaneous sign reversal preserves their joint distribution and sends the minimum to the negative maximum. Therefore $\mathbb{E}m_k = -\mathbb{E}M_k$. These expectations exist because $\max_j |Z_j| \le \sum_j |Z_j|$, whose expectation is finite. Writing $a_k = \mathbb{E}M_k$,

$$\boxed{\mathbb{E}R_k = 2\sigma\sqrt{1-\rho}\, a_k.} \tag{4.2}$$

The normal-range constant $2a_k$ is established statistical machinery. NIST uses the conventional notation $d_2(k)$ for the mean range of a normal sample divided by its SD (NIST/SEMATECH, n.d.). No novelty is claimed for this normal order-statistic result.

4.2 The expected independent-normal maximum

Independence gives

$$\mathbb{P}(M_k \le z) = \mathbb{P}(Z_1 \le z, \dots, Z_k \le z) = \Phi(z)^k.$$

Differentiating, $f_{M_k}(z) = k\phi(z)\Phi(z)^{k-1}$, and hence

$$a_k = k\int_{-\infty}^{\infty} z\phi(z)\Phi(z)^{k-1}\,\mathrm{d}z. \tag{4.3}$$

This is an exact integral, not the approximation obtained by evaluating a normal quantile at a plotting-position formula. For five tests its value is $a_5 = 1.16296447364052\ldots$. The next sections derive its closed form.

4.3 A Gaussian sign-probability identity

Lemma 4.1 (Two and three Gaussian signs)**.** *For a centered, standardized, jointly normal pair with correlation r,*

$$\mathbb{P}(V_1 > 0, V_2 > 0) = \frac{1}{4} + \frac{\arcsin r}{2\pi}.$$

For a centered jointly normal triple with nonzero marginal variances,

$$\mathbb{P}(V_1 > 0, V_2 > 0, V_3 > 0) = \frac{1}{8} + \frac{\arcsin r_{12} + \arcsin r_{13} + \arcsin r_{23}}{4\pi}. \tag{4.4}$$

Both identities are explicitly recorded in the introduction of Pinasco et al. (2021). An elementary proof is included to make the main derivation self-contained.

Proof. Represent the pair as A and $rA+\sqrt{1-r^2}B$, with A, B independent standard normals. The planar direction of (A, B) is uniform. The intersection of the two positive half-planes has angle $\pi - \arccos r = \pi/2 + \arcsin r$. Dividing by 2π gives the pair formula; the boundary correlations follow by continuity.

For the triple, put $S_j = \operatorname{sign}(V_j)$. The event indicator is $\prod_{j=1}^{3}(1+S_j)/8$, except on a null set. Central symmetry makes the expectation of every odd product of signs zero. The pair identity implies $\mathbb{E}(S_i S_j) = 2\arcsin(r_{ij})/\pi$. Expanding the product and taking its expectation yields equation (4.4). □

4.4 Closed form for five tests

Since $\phi'(z) = -z\phi(z)$, integration by parts in equation (4.3) gives

$$\begin{aligned} a_5 &= 5\int_{-\infty}^{\infty} z\phi(z)\Phi(z)^4 \,\mathrm{d}z \\ &= 20\int_{-\infty}^{\infty} \phi(z)^2\Phi(z)^3 \,\mathrm{d}z. \end{aligned} \tag{4.5}$$

The boundary term vanishes because $\Phi(z)^4$ is bounded and $\phi(z) \to 0$ at both infinities.

For $Y \sim N(0, 1/2)$, the density is $f_Y(y) = e^{-y^2}/\sqrt{\pi}$, so $\phi(y)^2 = f_Y(y)/(2\sqrt{\pi})$. Consequently

$$a_5 = \frac{10}{\sqrt{\pi}}\mathbb{E}\{\Phi(Y)^3\}.$$

Introduce three independent standard normals U_1, U_2, U_3, independent of Y. Conditioning on Y shows that

$$\mathbb{E}\{\Phi(Y)^3\} = \mathbb{P}(Y - U_1 > 0, Y - U_2 > 0, Y - U_3 > 0).$$

Each difference has variance $3/2$; two distinct differences have covariance $1/2$ and therefore correlation $1/3$. Applying equation (4.4),

$$\begin{aligned} a_5 &= \frac{10}{\sqrt{\pi}}\left\{\frac{1}{8} + \frac{3}{4\pi}\arcsin\frac{1}{3}\right\} \\ &= \boxed{\frac{5}{4\sqrt{\pi}}\left(1 + \frac{6}{\pi}\arcsin\frac{1}{3}\right)}. \end{aligned} \tag{4.6}$$

Theorem 4.2 (Five-test expected range)**.** *For equation (2.3) with $k = 5$ and $-1/4 \le \rho \le 1$,*

$$\boxed{\mathbb{E}R_5 = \frac{5\sigma\sqrt{1-\rho}}{2\sqrt{\pi}}\left(1 + \frac{6}{\pi}\arcsin\frac{1}{3}\right).} \tag{4.7}$$

At $\sigma = 15$, $\rho = 0.6$, this is $22.0656994473774\ldots$ points.

Proof. Substitute equation (4.6) into equation (4.2). □

An independent literature check is available in Finch (2016, section 1). His expected iid five-normal range is $10(1 - 3S_2)/\sqrt{\pi}$, where $S_2 = \arccos(1/3)/(2\pi)$. Using $\arccos(1/3) = \pi/2 - \arcsin(1/3)$ converts that expression to $2a_5$. The agreement identifies the result as an application of known range moments.

4.5 Highest and lowest scores

The same model gives

$$\mathbb{E}\max_j X_j = \mu + sa_5, \qquad \mathbb{E}\min_j X_j = \mu - sa_5.$$

At mean 100, these are 111.03285 and 88.96715. They are averages over the population of people taking all five tests. They do not predict the extrema of a particular person whose ability or previous test score is already known.

5 Individual variability and exceedance probabilities

5.1 Two-test difference distribution

For $d \geq 0$,

$$\mathbb{P}(A \leq d) = 2\Phi(d/\tau) - 1, \qquad Q_A(p) = \tau\Phi^{-1}\left(\frac{1+p}{2}\right). \tag{5.1}$$

Also $\mathbb{P}(A > d) = 2\{1 - \Phi(d/\tau)\}$. The centered signed difference lies between approximately -26.30 and 26.30 points with probability 0.95. That statement is equivalent to the 95th percentile of A being 26.30; it is not the same as the central 95% interval for the nonnegative variable A.

5.2 The exact range CDF

For $w > 0$, exactly one of the independent continuous Z_j is the minimum, almost surely. If a specified minimum is at z, the remaining $k-1$ scores must all fall in $[z, z+w]$ for the range to be at most w. Summing over which coordinate is the minimum gives

$$G_k(w) := \mathbb{P}(W_k \leq w) = k\int_{-\infty}^{\infty} \phi(z)[\Phi(z+w) - \Phi(z)]^{k-1}\,\mathrm{d}z. \tag{5.2}$$

For $r \geq 0$ and $s > 0$, $\mathbb{P}(R_k \leq r) = G_k(r/s)$. At $s = 0$, the range is identically zero.

R's `ptukey` and `qtukey` describe a studentized normal range. With infinite degrees of freedom, the independent scale denominator is one, so they compute this unstudentized range CDF and its quantiles (R Core Team, 2026d). No finite-sample estimate of a variance is being substituted here. The supplement independently integrates equation (5.2) and inverts it, checking this use of the R functions.

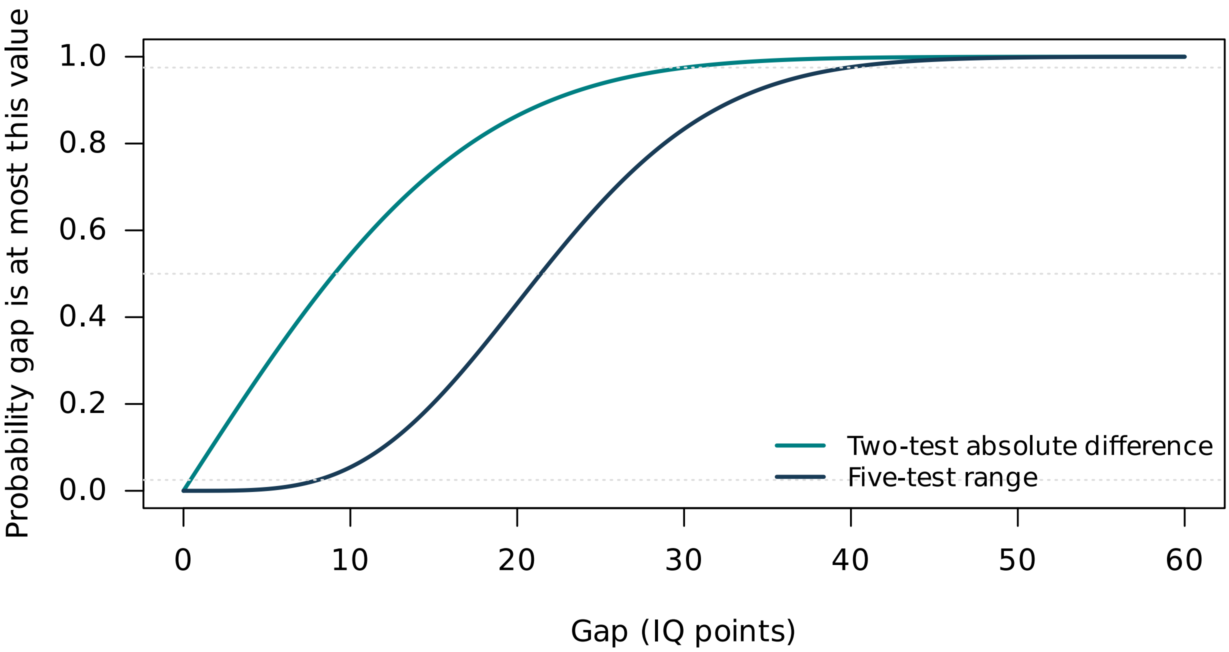

Figure 5.1: Model CDFs at SD 15 and correlation 0.6. Each curve gives the fraction of people whose gap is at most the horizontal-axis value. Curves are theoretical, not fitted to observations.

The 2.5th and 97.5th percentiles delimit the central 95% interval. A threshold chosen to be exceeded by only 5% of people instead uses the 95th percentile: 36.60 points for the five-test range. Neither interval is a confidence interval for a true IQ or an estimated mean.

Table 5.1: Selected theoretical quantiles, in IQ points.

Probability	Two-test absolute gap	Five-test range
0.025	0.420	8.061
0.500	9.049	21.411
0.950	26.296	36.597
0.975	30.072	39.816

Table 5.2: Percentage exceeding a specified gap under the main model.

Threshold (points)	Absolute gap (%)	Five-test range (%)
5	70.94	99.59
10	45.61	94.58
15	26.36	79.71
20	13.60	56.85
30	2.53	16.64
40	0.29	2.40

The range exceeds 20 points for approximately 56.85% of people, although one specified pair differs by more than 20 points for only 13.60%. There are ten opportunities for a pairwise gap in a set of five, but they are dependent. The range distribution accounts for that dependence exactly. Multiplying a single-pair probability by ten, or treating all ten pairs as independent, would not give the exact answer.

5.3 A second-moment check

Finch (2016, section 1) also provides the iid five-normal range's second raw moment, which in elementary inverse-trigonometric notation is

$$\nu_5 = \mathbb{E}W_5^2 = 2\left[1 + \frac{5\sqrt{3}}{2\pi} + \frac{15}{\pi^2}\arccos\frac{2}{3} - \frac{5\sqrt{3}}{2\pi^2}\arccos\frac{1}{4}\right] = 6.15658306873\ldots. \tag{5.3}$$

Unlike the mean formula, this second-moment identity is imported from the cited literature rather than proved here. Since $\mathbb{E}W_5 = 2a_5$,

$$\mathrm{SD}(R_5) = s\sqrt{\nu_5 - (2a_5)^2} = 8.19740105668\ldots.$$

The exact mean's Monte Carlo SE at five million independent people is $8.19740105668/\sqrt{5{,}000{,}000} = 0.0036659892$ points. This is a separate check on the simulation's estimated SE of 0.0036653854.

6 What the assumptions do: three counterexamples

6.1 Normal marginals and the complete correlation matrix are not enough

A jointly normal law is much more restrictive than a list of normal marginal distributions. Dutta and Genton (2014) construct non-Gaussian multivariate laws even with all proper subvectors Gaussian. The simpler constructions below are proved directly and are sufficient to refute identification from marginal normality and correlations alone.

Let B be Bernoulli with success probability ρ, independent of independent standard normals $Z_0, Z_1, \dots, Z_k$. Define

$$\widetilde{X}_j = \begin{cases} \mu + \sigma Z_0, & B = 1, \\ \mu + \sigma Z_j, & B = 0. \end{cases} \tag{6.1}$$

Both conditional marginal laws are $N(\mu, \sigma^2)$, so the unconditional marginal is exactly the same normal law. For $i \neq j$,

$$\mathrm{Cov}(\widetilde{X}_i, \widetilde{X}_j) = \rho\sigma^2 + (1-\rho)0 = \rho\sigma^2.$$

Thus every pair has correlation ρ. When $B = 1$, the range and absolute gap are zero. When $B = 0$, the scores are independent. It follows that

$$\mathbb{E}|\widetilde{X}_1 - \widetilde{X}_2| = (1-\rho)\frac{2\sigma}{\sqrt{\pi}}, \qquad \mathbb{E}\widetilde{R}_k = (1-\rho)2\sigma a_k. \tag{6.2}$$

At $\rho = 0.6$ these expectations are 6.7702750 and 13.9555737 points. The construction puts probability 0.6 on exactly identical scores. That makes it an especially transparent proof, but is not intended as a realistic IQ model.

6.2 A smooth counterexample without exact ties

The failure of identification is not an artefact of that point mass. Let a latent variable C take the values 0.3 and 0.9 with equal probability, and conditionally on $C = c$ let

$$X_j = \mu + \sigma\sqrt{c}\, Z_0 + \sigma\sqrt{1-c}\, Z_j.$$

Each conditional marginal is $N(\mu, \sigma^2)$, irrespective of c, so every unconditional marginal is normal. The covariance of any distinct pair is $\sigma^2 \mathbb{E}C = 0.6\sigma^2$. Both component covariance matrices are positive definite; the mixture has a smooth, positive density, and exact ties have probability zero.

Conditional expectation gives

$$\mathbb{E}|X_1 - X_2| = \frac{2\sigma}{\sqrt{\pi}}\mathbb{E}\sqrt{1-C},$$
$$\mathbb{E}R_5 = 2\sigma a_5 \mathbb{E}\sqrt{1-C},$$

where $\mathbb{E}\sqrt{1-C} = (\sqrt{0.7} + \sqrt{0.1})/2$. This yields 9.7567093 and 20.1115132 points. Strict concavity of the square root gives $\mathbb{E}\sqrt{1-C} < \sqrt{1-\mathbb{E}C}$ for nonconstant C, explaining why these means are smaller than those of the single Gaussian model with the same overall correlation. This comparison is specific to this mixture family; it does not assert that the Gaussian law maximizes these expectations among all possible laws with these moments.

The main joint-normal assumption is sufficient, not logically necessary, for the main numeric answers. Another law could happen to share them. What fails is the claim that normal marginals and correlation force those answers.

Table 6.1: Same normal marginals, same full correlation matrix, different answers.

Joint law	Mean absolute gap	Mean range
Single Gaussian	10.7047	22.0657
Shared/independent mixture	6.7703	13.9556
Smooth Gaussian mixture	9.7567	20.1115

6.3 Even a Gaussian average correlation is not enough

Let (A, B) be jointly normal, with common mean μ, SD σ, and correlation $1/3$. Consider the five-vector

$$\mathbf{X} = (A, A, A, B, B).$$

There are four within-block pairs with correlation one and six between-block pairs with correlation $1/3$. Their average correlation is

$$\frac{4 \cdot 1 + 6 \cdot (1/3)}{10} = 0.6.$$

Yet the five-test range is simply $|A - B|$, so

$$\mathbb{E}R_5 = 2\sigma\sqrt{\frac{2/3}{\pi}} = 13.8197659789\ldots \text{ points at } \sigma = 15.$$

This is a singular Gaussian counterexample. It meets the weaker claim "average correlation 0.6," not the stronger assumption that each pair has correlation 0.6. Singularity is admissible for a probability counterexample; the point is that an average discards information needed by a nonlinear statistic such as the maximum-minus-minimum.

6.4 Useful bounds from moments alone

Without joint normality, Cauchy–Schwarz still gives

$$\mathbb{E}|D| \leq \sqrt{\mathbb{E}D^2} = \sigma\sqrt{2(1-\rho)}. \tag{6.3}$$

For equal means, variances, and equicorrelation, a range bound also follows. Let $\bar{X} = k^{-1}\sum_j X_j$. For every realized vector,

$$R_k^2 \leq 2\{(\max_j X_j - \bar{X})^2 + (\min_j X_j - \bar{X})^2\} \leq 2\sum_j (X_j - \bar{X})^2.$$

The expectation of the last sum is $k\sigma^2 - k\,\mathrm{Var}(\bar{X}) = (k-1)\sigma^2(1-\rho)$. Therefore

$$\mathbb{E}R_k \leq \sqrt{\mathbb{E}R_k^2} \leq \sigma\sqrt{2(k-1)(1-\rho)}. \tag{6.4}$$

At the main parameters, these upper bounds are 13.4164 points for the absolute gap and 26.8328 points for the five-test range. They are bounds under the moment assumptions, not alternative Gaussian expectations. No claim of sharpness is made for the specified normal-marginal class. General moment-based bounds for order statistics are studied by Bertsimas et al. (2006); the elementary inequalities above are derived here.

7 Conditioning and sensitivity

7.1 After a first score has been observed

The question "How far apart are two scores on average?" differs from "What should I expect after obtaining a score of 145?" The conditional normal formula (Soch, 2020) gives, under the main bivariate model,

$$X_2 \mid X_1 = x \sim N(\mu + \rho(x-\mu), \sigma^2(1-\rho^2))\,. \tag{7.1}$$

Thus $D \mid X_1 = x$ has mean $\delta_x = (1-\rho)(x-\mu)$ and SD $\tau_c = \sigma\sqrt{1-\rho^2}$. Its expected absolute value is equation (3.3) with those parameters. Here $\tau_c = 12$ points.

Table 7.1: Conditional predictions at mean 100, SD 15, and correlation 0.6.

First score	Mean second score	Mean signed gap	Mean absolute gap
85	91	-6	10.747
100	100	0	9.575
115	109	6	10.747
130	118	12	14.000
145	127	18	18.703
160	136	24	24.204

The conditional shift toward the population mean is regression to the mean in this statistical model. It is not evidence of a causal decline in a person's ability. The extreme-score rows also extrapolate the assumed normal model into a region where real-test ceilings and calibration may matter.

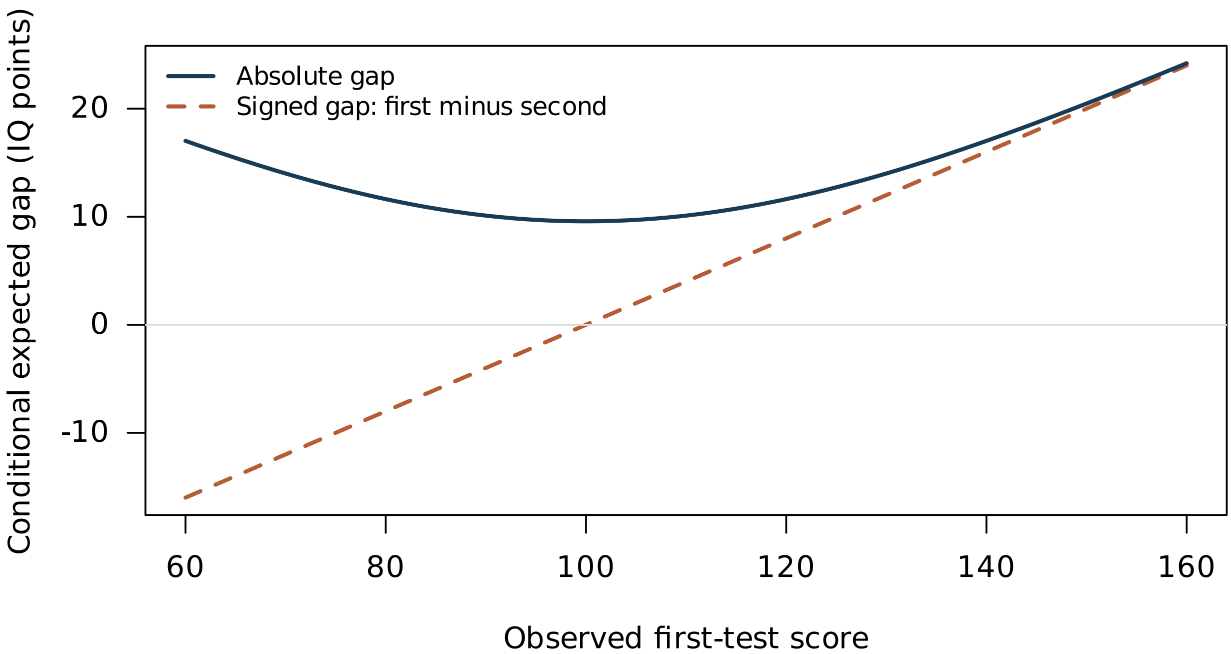

Figure 7.1: Conditional expected signed and absolute gaps after observing the first score. The signed gap is first minus second. The model is symmetric about 100 for the absolute gap.

For all remaining tests $j \geq 2$, conditioning on $X_1 = x$ gives means $\mu + \rho(x-\mu)$, variances $\sigma^2(1-\rho^2)$, and off-diagonal covariances $\sigma^2\rho(1-\rho)$. Their conditional correlations are $\rho/(1+\rho)$, or 0.375 here. The first score is now fixed and the remaining four are still dependent. Consequently the unconditional five-test range distribution cannot simply be reused for a five-test range conditional on that first score.

7.2 Selecting high scorers is different again

Selecting everyone with $X_1 > c$ is not equivalent to conditioning at $X_1 = c$. If $z = (c-\mu)/\sigma$, then the truncated-normal identity $\int_z^\infty u\phi(u)\,\mathrm{d}u = \phi(z)$ gives

$$\mathbb{E}(X_1 - X_2 \mid X_1 > c) = (1-\rho)\sigma\frac{\phi(z)}{1-\Phi(z)}.$$

This is another population average with a different inclusion rule. Membership in a group selected on any of several scores, self-selection into testing, and publication of only the best score create different conditioning events again. The present report does not estimate those selection mechanisms.

7.3 The range and the person's average score

An instructive feature of the exchangeable Gaussian model is that $\bar{X}$ is independent of the centered vector $(X_j - \bar{X})_{j=1}^k$. Indeed, $\mathrm{Cov}(\bar{X}, X_j - \bar{X}) = 0$ for every j, and joint normality converts this zero covariance into independence. Since the range is a function of that centered vector, R_k is independent of $\bar{X}$ under this exact model. Conditioning on the *average of all five* is therefore different from conditioning on the *first score*. This mathematical property should not be presumed for real scores with unequal covariance structures.

7.4 Changing the correlation, scale, or number of tests

Both requested expectations scale linearly with σ and, under the main model, proportionally to $\sqrt{1-\rho}$. For $f(\rho) = C\sqrt{1-\rho}$,

$$f'(\rho) = -\frac{C}{2\sqrt{1-\rho}}, \qquad \frac{f'(\rho)}{f(\rho)} = -\frac{1}{2(1-\rho)}.$$

Near 0.6, a correlation change of 0.05 corresponds to approximately a 6.25% oppositely directed change in either expectation. This is a local sensitivity, not an uncertainty interval for a correlation estimate.

Table 7.2: Sensitivity to the common pairwise correlation, with SD 15.

Correlation	Mean absolute gap	Mean five-test range
0.0	16.926	34.889
0.3	14.161	29.190
0.5	11.968	24.670
0.6	10.705	22.066
0.7	9.271	19.109
0.8	7.569	15.603
0.9	5.352	11.033
1.0	0.000	0.000

For a different common SD, multiply every point-valued answer by $\sigma/15$. For example, SD 16 gives 11.4184 and 23.5367 points. Adding more tests increases the range pointwise for nested test sets, whereas the expected absolute gap for a specified pair stays unchanged.

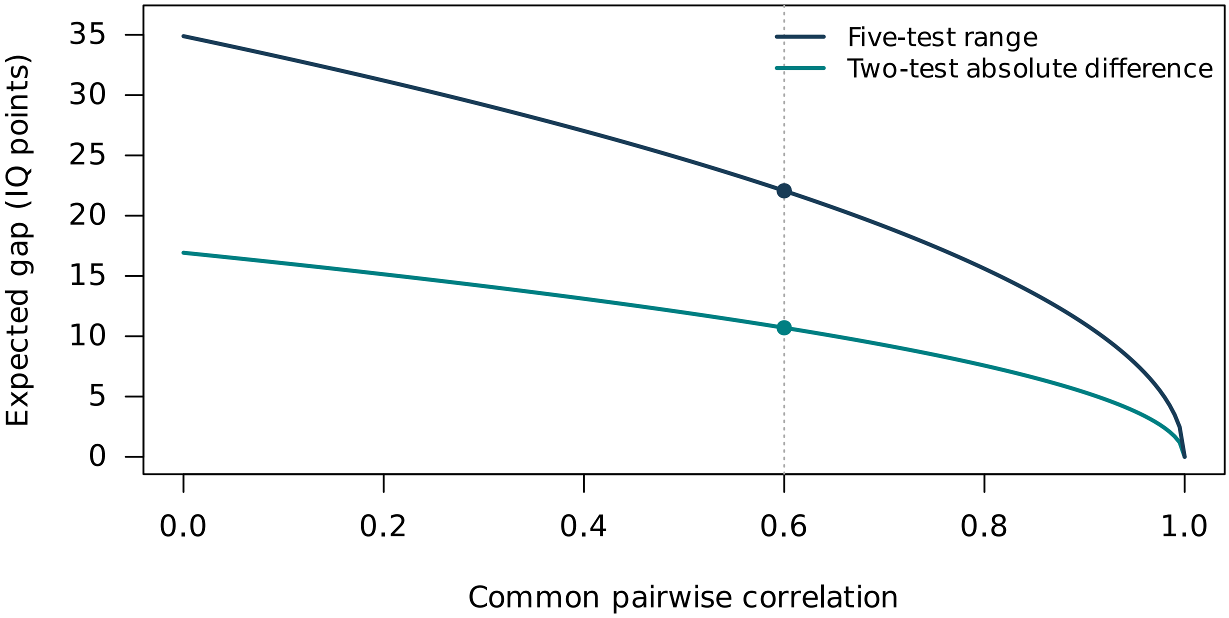

Figure 7.2: Correlation sensitivity at fixed SD 15. Both expectations shrink by $\sqrt{1-\rho}$; the dots mark $\rho = 0.6$.

Table 7.3: Expected ranges for different numbers of equicorrelated tests, at SD 15 and $\rho = 0.6$.

Tests k	$a_k = \mathbb{E}\max_j Z_j$	**Mean range (points)**
2	0.564190	10.705
3	0.846284	16.057
4	1.029375	19.531
5	1.162964	22.066
10	1.538753	29.196
20	1.867475	35.433
50	2.249074	42.673

8 Psychometric interpretation and limits

8.1 Intertest correlation is not automatically reliability

The correlation in this report is the correlation between observed scores across the specified population. Reliability concerns consistency over specified replications of a measurement procedure. The American Educational Research Association et al. (2014) distinguish internal consistency, alternate-form, and test–retest approaches; Standard 2.4 specifically addresses precision of interpreted score differences.

Under an additional parallel-test model $X_j = T + \varepsilon_j$, with one shared true score, independent zero-mean errors, and equal error variances, the intertest correlation can equal $\operatorname{Var}(T)/\operatorname{Var}(X_j)$. In that special model, $\sigma\sqrt{1-\rho}$ is a standard error of measurement. Without those assumptions, test-specific stable abilities, content differences, or other nonshared influences can also contribute. The factor representation in equation (2.4) alone does not distinguish these possibilities.

For illustration, a Pearson technical report gives average reliability 0.95 for two particular WISC-V expanded indexes and 0.96 for FSIQ (Raiford et al., 2015). These publisher coefficients are neither correlations between five distinct IQ tests nor substitutes for the stipulated 0.6. Using a high internal reliability coefficient in an intertest discrepancy formula without checking its interpretation can answer the wrong question.

8.2 Agreement, discrepancy base rates, and population spread

Bland and Altman (1986) explain why a high Pearson correlation need not indicate close numerical agreement. Their analysis also highlights dependence on the spread of the sampled population: correlations from a broad population need not transfer unchanged to a restricted group. The Gaussian difference distribution derived here is a population agreement calculation under equal means and variances. It does not identify clinically or educationally important discrepancies.

Pearson's score documentation distinguishes the statistical significance of a discrepancy from its base rate in a normative sample (NCS Pearson, 2020; Raiford et al., 2015). The probabilities in table 5.2 are *model-based* base rates for an idealized set of tests; they are not publisher-provided norms for a particular battery. In particular, a five-test range of 30 points has probability approximately 16.64% of being exceeded under this model, so that gap alone is not exceptionally rare in the modeled population.

Standard 2.20 in American Educational Research Association et al. (2014) addresses disclosure of adjustments for range restriction. For this report, no adjustment can be calculated because there is no actual sample or selection rule. The appropriate response is to identify the population to which a correlation belongs before applying it.

8.3 Practice and retest effects are real additional mechanisms

Empirical studies show why occasion-specific means cannot always be assumed equal. Three relevant examples were checked at abstract level:

- Hausknecht et al. (2007) synthesized 50 studies with 107 samples and 134,436 participants in cognitive selection testing. Their adjusted overall retest effect size was 0.26, with larger gains for coaching and identical forms. The heterogeneous selection contexts do not justify a universal IQ-point correction.
- Estevis et al. (2012) studied 54 young adults, with mean age 20.9 and initial mean FSIQ 111.6, retested on WAIS-IV after three or six months. The abstract reports an approximately seven-point FSIQ gain. The small, young, above-average sample and same-test design limit generalization.

- Scharfen et al. (2018) synthesized 174 samples from 122 studies, totaling 153,185 participants. The abstract reports retest effects moderated by test content/operation, form equivalence, interval, and age, with no further aggregate gains after the third administration in that synthesis. That aggregate pattern is not a guarantee for a given individual or test.

These findings motivate more realistic models; they do not alter the arithmetic of the hypothetical equal-mean Gaussian model. No detailed moderator estimates or effect-size tables from unavailable full texts are used.

8.4 Rounding, ceilings, and heterogeneous tests

The normal model is unbounded. Actual reported scores may be discrete and bounded; the WISC-V sample report explicitly documents bounded index and FSIQ ranges (NCS Pearson, 2020). Such reporting rules can matter particularly near the tails.

A simple deterministic rounding bound is available. If every score is rounded to the nearest unit, each changes by at most 0.5 point. The absolute gap and the range each change by at most one point, by the triangle inequality and the fact that a maximum or minimum changes by at most the largest coordinate perturbation. Thus their expectations change by at most one point under a coupling to the same unrounded scores. This bound is conservative; it is not an estimate of the actual rounding effect. The rounded scores' correlations may also differ from the latent continuous correlations.

For known unequal means, variances, and a full covariance matrix Σ, a multivariate-normal simulation can use a matrix factor L satisfying $LL^\top = \Sigma$ and generate $\boldsymbol{\mu} + L\mathbf{Z}$. The five-range simplification does not generally reduce to a single $\sqrt{1-\rho}$ factor. Floor/ceiling rules, practice effects, or selection should be represented explicitly if they are part of the intended estimand.

9 Simulation design and actual results

9.1 What the simulation is intended to establish

The design specifies the aim, data-generating mechanisms, target expectations, calculation methods, and checks. This follows the planning logic of the ADEMP framework described by Morris et al. (2019). Here the experiment checks known probability results, rather than comparing competing estimators on an unknown empirical truth.

The main program uses base R only. It simulates 5,000,000 independent people per model and five correlated scores per person, in batches of 100,000. For each person it calculates one prespecified pair's signed, absolute, and squared differences, plus the full range, maximum, and minimum. It also accumulates all score means and covariances as a check on the data generator.

`rnorm` supplies the normal draws (R Core Team, 2026b). The generator is explicitly set to Mersenne-Twister with inversion for normal variates and rejection for sampling; the main seed is 20260903 (R Core Team, 2026c). The shared/independent mixture continues the random stream after the Gaussian run. Changing batch size changes draw ordering and can change the realized sample, although not its target law. The seed alone is not a complete specification of a run; script version, parameters, RNG kinds, R version, and batch size also matter.

9.2 Monte Carlo uncertainty

If T_i is one person's target statistic, then

$$\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} T_i, \qquad \widehat{\mathrm{MCSE}}(\hat{\theta}) = \frac{s_T}{\sqrt{n}}.$$

The code uses batch means and centered sums of squares, merging them using the between-batch correction. This avoids subtracting two nearly equal large raw second-moment accumulations. The approximate Monte Carlo interval is $\hat{\theta} \pm 1.96\,\widehat{\mathrm{MCSE}}$.

The strong law applies because the person-level statistics are iid with finite expectations. The central limit theorem applies to these sample means because their variances are finite. In these normal and finite-normal-mixture models, all the moments used exist. Thus Monte Carlo averages converge to the proved model expectations. A finite run does not prove an identity or validate a model for real IQ tests.

Table 9.1: Fresh main R run: 5,000,000 people per model. Differences and ranges are in IQ points.

Statistic	Theory	Simulated	MC SE	Error / SE
Gaussian signed gap	0.000000	0.011731	0.005999	1.955
Gaussian absolute gap	10.704745	10.701215	0.003617	-0.976
Gaussian five-test range	22.065699	22.063505	0.003665	-0.599
Mixture absolute gap	6.770275	6.767670	0.005180	-0.503
Mixture five-test range	13.955574	13.947077	0.008474	-1.003

Both requested Gaussian expectations are within one estimated MC SE of theory. The Gaussian mean signed gap is about 1.96 SE from zero, an ordinary finite-sample fluctuation; its approximate 95% Monte Carlo interval is $[-0.0000269, 0.0234885]$ points. It is not evidence of a systematic test bias.

The ten empirical Gaussian pair correlations range from 0.599650 to 0.600389. Marginal test means range from approximately 99.99745 to 100.01079, and SDs from 14.99656 to 15.00310. These are checks on a finite simulated sample, not exact constraints imposed on that sample.

The newly executed main log is byte-for-byte identical to the earlier saved log supplied by the preceding analysis. That verifies reproduction of the reported run in this environment. It is not an independent random replication: the seed and algorithm were intentionally held fixed.

9.3 Separate smooth-mixture experiment

The supplement runs 1,000,000 people, using seed 20260904, for the continuous two-component model in chapter 6.

Statistic	Theory	Simulated	MC SE
Absolute gap	9.756709	9.760979	0.009218
Five-test range	20.111513	20.103200	0.012223

Both deviations are below one MC SE. This is an actual run of the supplied code, not a fabricated table of values chosen to match the formulas.

9.4 Deterministic numerical checks

1. Adaptive quadrature of equation (4.3) agrees with the exact five-normal formula to displayed precision; the reported integration error estimate is about 4.21×10^{-12}. R's `integrate` uses adaptive quadrature and supports infinite limits (R Core Team, 2026a). Its error estimate is numerical, not a formal mathematical enclosure.
2. The two-normal integral equals $1/\sqrt{\pi}$, checking that the general range result reduces to the two-test absolute-gap formula.
3. The independently coded CDF integral agrees with `ptukey` at six test arguments, with maximum observed absolute difference about 4.28×10^{-12}. Inverting that integral independently confirms the reported range quantiles; the 97.5th-percentile difference is about 0.000003 IQ point.
4. The factor-loadings covariance matches the target covariance matrix. The supplement also checks the projection construction at correlations $-0.25, -0.1, 0, 0.6, 1$.
5. The exact second-moment calculation predicts the Gaussian range MC SE independently of the simulated variance estimate.

The simulation diagnostic flags deviations larger than six estimated SEs; it is not a hypothesis test or a rule to keep changing seeds until a result looks satisfactory. No such resampling was used to select the reported results.

10 Evidence, conclusions, and unresolved questions

10.1 Research approach and source quality

Research was conducted on 3 September 2026 using targeted public web searches and direct source reads. The search families were: normal order statistics and five-normal range moments; Gaussian orthant probabilities; marginal versus joint normality; equicorrelation matrices and moment bounds; correlation versus agreement; psychometric reliability and discrepancy interpretation; cognitive retest effects; and official R documentation for simulation and integration.

Examples of actual search targets included “normal range expected maximum five arcsin”, “orthant probabilities Pinasco Smucler Zalduendo”, “Bland Altman agreement correlation”, “retest effects cognitive ability meta-analysis”, and site-restricted searches for R’s Tukey distribution and random-number-generation documentation. This was a focused research synthesis, not a registered systematic review or an exhaustive search of every database.

Primary papers, author-hosted manuscripts, official test-publisher documents, the official testing standards, institutional documentation, and authored probability proofs were prioritized. Research into mathematical foundations and psychometric interpretation was divided between two additional AI research workers; the coordinating assistant reconciled their findings, revisited key sources, and checked the main calculations. These were not independent human reviewers.

10.2 What was verified, and at what level

Table 10.1: Evidence map and limits of access.

Claim family	Main evidence	Scope or limitation
Exact range mean	Finch; orthant identity in Pinasco et al.; proof in this report	Full relevant preprint passages checked. Formula is established, not a claimed new discovery.
Joint normality	Pishro-Nik; Taboga; Dutta and Genton	Full relevant sections checked; counterexamples in this report proved directly.
Covariance constraints	Archakov et al.; direct eigenvalue calculation	Relevant result read in a 2022 preliminary version; published metadata is from 2024.
Moment-only bounds	Bertsimas et al.; elementary bounds proved here	Full paper available. No claim that this report solves the sharp optimization problem.
Score scales and precision	Pearson sample/technical reports; 2014 Testing Standards	Relevant official sections read. Specific batteries do not validate the hypothetical five-test model.
Practice effects	Hausknecht et al.; Estevis et al.; Scharfen et al.	Original abstracts or author-associated abstract record only; full study methods and detailed tables not reviewed.

Continued on next page

Table 10.1: Evidence map and limits of access. (Continued)

Claim family	Main evidence	Scope or limitation
Simulation design	Morris et al.; official R manuals; actual R logs	Repository/abstract framework checked. Online R development docs and actual R 4.3.3 are distinguished.
Actual five-test applicability	None supplied	No named tests, real score matrix, or estimated target-population covariance. This remains unverified.

The most consequential mathematical claim was checked through a self-contained derivation, an independently tabulated literature formula, numerical integration, and R simulation. The exact second-moment formula was taken from a cited source and checked numerically, but not independently rederived. Searches stopped when the substantive sections had adequate evidence and the remaining gap was lack of real test data rather than lack of further general references.

10.3 What a real-data extension would need

To make an empirical prediction for particular tests, one would need their identities and score scales, the target population and sampling/selection rule, the full score matrix or a defensible covariance estimate, and administration order, timing, prior exposure, and ceiling/floor information. Those data would allow checks of mean/variance equality, covariance heterogeneity, discrepancy distributions, tail behavior, and the effect of conditioning on an observed score. If sampling uncertainty in the covariance were material, it would need to be propagated into predictions, separately from Monte Carlo error.

Supported conclusion

For the explicitly assumed exchangeable joint-normal model, the requested averages are rigorously established: **10.70 IQ points** for a two-test absolute gap and **22.07 IQ points** for a five-test range. Correlation 0.6 alone is insufficient, even with normal marginal score distributions. The results are a mathematical benchmark whose empirical applicability must be checked for the particular tests and population.

References

American Educational Research Association, American Psychological Association, and National Council on Measurement in Education. *Standards for Educational and Psychological Testing*. American Educational Research Association, 2014. URL https://www.testingstandards.net/uploads/7/6/6/4/76643089/standards_2014edition.pdf. Chapter 2, especially Standards 2.4, 2.6 and 2.20. This report cites the 2014 edition.

Ilya Archakov, Peter Reinhard Hansen, and Yiyao Luo. A new method for generating random correlation matrices. *The Econometrics Journal*, 27(2):188–212, 2024. doi: 10.1093/ectj/utad027. URL https://econ.la.psu.edu/wp-content/uploads/sites/5/2022/01/GenRanCorr.pdf. Equicorrelation result read in the public preliminary version dated 27 February 2022, section 2.4.

Dimitris Bertsimas, Karthik Natarajan, and Chung-Piaw Teo. Tight bounds on expected order statistics. *Probability in the Engineering and Informational Sciences*, 20(4):667–686, 2006. doi: 10.1017/S0269964806060414. URL https://www.mit.edu/~dbertsim/papers/MomentProblems/Tight%20bounds%20on%20expected%20order%20statistics.pdf.

J. Martin Bland and Douglas G. Altman. Statistical methods for assessing agreement between two methods of clinical measurement. *The Lancet*, 1(8476):307–310, 1986. doi: 10.1016/S0140-6736(86)90837-8. URL https://www-users.york.ac.uk/~mb55/meas/ba.htm. Author-hosted full text consulted.

Subhajit Dutta and Marc G. Genton. A non-Gaussian multivariate distribution with all lower-dimensional Gaussians and related families. *Journal of Multivariate Analysis*, 132:82–93, 2014. doi: 10.1016/j.jmva.2014.07.007. URL https://marcgenton.github.io/2014.DG.JMVA.pdf.

Eduardo Estevis, Michael R. Basso, and Dennis Combs. Effects of practice on the Wechsler Adult Intelligence Scale-IV across 3- and 6-month intervals. *The Clinical Neuropsychologist*, 26(2):239–254, 2012. doi: 10.1080/13854046.2012.659219. URL https://pubmed.ncbi.nlm.nih.gov/22353021/. Original abstract and bibliographic record consulted; full article not reviewed.

Steven R. Finch. Mean width of a regular simplex. arXiv:1111.4976v2, 2016. URL https://arxiv.org/pdf/1111.4976. Version dated 12 March 2016; section 1, p. 2 consulted.

John P. Hausknecht, Jonathan A. Halpert, Nicole T. Di Paolo, and Mary O. Moriarty Gerrard. Retesting in selection: A meta-analysis of coaching and practice effects for tests of cognitive ability. *Journal of Applied Psychology*, 92(2):373–385, 2007. doi: 10.1037/0021-9010.92.2.373. URL https://pubmed.ncbi.nlm.nih.gov/17371085/. Original abstract and bibliographic record consulted; full article not reviewed.

Tim P. Morris, Ian R. White, and Michael J. Crowther. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, 38:2074–2102, 2019. doi: 10.1002/sim.8086. URL https://figshare.le.ac.uk/articles/journal_contribution/Using_simulation_studies_to_evaluate_statistical_methods_/10230053. University repository and public abstract/preprint metadata consulted for simulation-design framework.

NCS Pearson. WISC-V interpretive report: Sample report, 2020. URL https://www.pearsonclinical.asia/content/dam/school/global/clinical/us/assets/wisc-v/wisc-v-interpretive-report.pdf. Report dated 20 October 2020, copyright 2015; score metric on PDF p. 3.

NIST/SEMATECH. Shewhart x-bar and r and s control charts. e-Handbook of Statistical Methods, section 6.3.2.1, n.d. URL https://www.itl.nist.gov/div898/handbook/pmc/section3/pmc321.htm. Normal-range constants and range-based scale estimation.

Damián Pinasco, Ezequiel Smucler, and Ignacio Zalduendo. Orthant probabilities and the attainment of maxima on a vertex of a simplex. *Linear Algebra and its Applications*, 610:785–803, 2021. URL https://arxiv.org/pdf/2004.04682. Public preprint dated 9 April 2020 consulted; orthant identities on p. 1.

Hossein Pishro-Nik. *Introduction to Probability, Statistics, and Random Processes*. Kappa Research, 2014. URL https://www.probabilitycourse.com/chapter5/5_3_2_bivariate_normal_dist.php. Open textbook, section 5.3.2 on bivariate normal distributions.

R Core Team. Integration of one-dimensional functions. R stats documentation, 2026a. URL https://stat.ethz.ch/R-manual/R-devel/library/stats/html/integrate.html. Adaptive quadrature documentation; actual computations used R 4.3.3.

R Core Team. The normal distribution. R stats documentation, 2026b. URL https://stat.ethz.ch/R-manual/R-devel/library/stats/html/Normal.html. Online development manual consulted; actual computations used R 4.3.3.

R Core Team. Random number generation. R base documentation, 2026c. URL https://stat.ethz.ch/R-manual/R-devel/library/base/html/Random.html. RNGkind and set.seed documentation; actual computations used R 4.3.3.

R Core Team. The studentized range distribution. R stats documentation, 2026d. URL https://stat.ethz.ch/R-manual/R-devel/library/stats/html/Tukey.html. Documentation for ptukey and qtukey; actual computations used R 4.3.3.

Susan Engi Raiford, Lisa Drozdick, Ou Zhang, and Xuechun Zhou. WISC-V technical report 1: Expanded index scores. Technical report, NCS Pearson, August 2015. URL https://www.pearsonclinical.in/content/dam/school/global/clinical/us/assets/wisc-v/wisc-v-technical-report-1.pdf. Relevant reliability and discrepancy sections, pp. 5–8.

Jana Scharfen, Judith Marie Peters, and Heinz Holling. Retest effects in cognitive ability tests: A meta-analysis. *Intelligence*, 67:44–66, 2018. doi: 10.1016/j.intell.2018.01.003. URL https://www.researchgate.net/publication/322681573_Retest_effects_in_cognitive_ability_tests_A_meta-analysis. Author-associated abstract record consulted; publisher full text unavailable.

Joram Soch. Conditional distributions of the multivariate normal distribution. The Book of Statistical Proofs, 2020. URL https://statproofbook.github.io/P/mvn-cond.html. Proof dated 20 March 2020.

Marco Taboga. Linear combinations of normal random variables. StatLect, n.d. URL https://www.statlect.com/probability-distributions/normal-distribution-linear-combinations.

A Reproducing the report

A.1 Files and computational environment

The source package contains the main XeLaTeX file, its complete body source, the BibTeX database, two R programs, the workbook export program, build commands, the actual run logs, vector figures, numeric CSV tables, and a companion workbook. The simulations use only base R; the reported runs used R 4.3.3 (2024-02-29) on 64-bit Linux. Exact session details are in the logs. The workbook is an export of model calculations and plotted values, not a dataset of measured IQ scores.

The source-code appendices print every analysis, simulation, figure-generation, workbook-export, and build program used in the deliverable. The XeLaTeX document and bibliography sources are supplied as editable files in the package rather than recursively printed inside their own output. External R, TeX, and workbook library implementations are dependencies, not newly authored report code.

A.2 Commands

The complete build script below compiles the supplied sources and figures. Its commented R commands can be run separately to reproduce the calculations:

```
#!/usr/bin/env bash
set -euo pipefail
# Run from the extracted report directory.
# Uses supplied figures and tables; does not overwrite simulation logs.
latexmk -xelatex -interaction=nonstopmode -halt-on-error iq_test_differences_report.tex
# To regenerate derived figures/tables first:
# Rscript code/iq_report_extensions.R
# To repeat the main simulation, choose a NEW output filename:
# Rscript code/iq_difference_simulation.R --out=results/main_run_new.txt
```

The first program refuses to overwrite an existing log. The supplement regenerates its own derived tables, figures, and results. The precomputed vector figures are included, so compiling the report alone does not require rerunning the simulations or exporting the workbook.

The requested TeX components are loaded explicitly: `scrbook`, `fontspec`, `unicode-math`, `tcolorbox`, `tabularray`, `hyperref`, `bookmark`, `mathtools`, `microtype`, `scrlayer-scrpage`, and `cleveref`. Additional packages provide geometry, figures, program listings, and author–year citations. Duplicate package names in the request are loaded once; the intended hyperlink package is spelled `hyperref`.

A.3 Numeric data and precision

The `tables` directory contains unrounded CSV values for the probability, quantile, sensitivity, conditional, and counterexample tables, plus every plotted curve. Calculations involving ordinary normal functions use R's double precision. R documents `qtukey` accuracy at about the fourth decimal place in its own units; the independent inversion check here agrees more closely at the reported probabilities. Main-text range quantiles are therefore rounded appropriately rather than presented as mathematically exact decimals.

The separate `iq_report_data.xlsx` workbook includes a formula-based parameter sheet and the computed table/curve values, with units and source notes. The exporter uses `@oai/artifact-tool`; it is optional for reproducing the R results or the PDF. The complete export source is printed for transparency, but users without that library can use the CSV files directly.

B Complete main R simulation source

The following program is the unchanged simulation script from the preceding analysis. It was executed again for this report. Its comments, input validation, deterministic checks, accumulators, both model generators, output formatting, and environment reporting are included in full.

```r
#!/usr/bin/env Rscript
# Expected IQ-score differences and ranges, with reproducible Monte Carlo checks.
# Base R only; no package installation is needed.
#
# Run:
#   Rscript iq_difference_simulation.R
#   Rscript iq_difference_simulation.R --n=1000000 --rho=0.6 --sd=15
#   Rscript iq_difference_simulation.R --k=10 --out=iq_10_test_results.txt
#
# Each row represents an independent person taking the same k tests.
# Gaussian model: X_j = mu + sigma*(sqrt(rho)*Z0 + sqrt(1-rho)*Zj).
# Z0,Z1,...,Zk are mutually independent standard normal variables.
# Thus every test has mean mu, SD sigma, and each pair has correlation rho.
#
# E|X1-X2| = 2*sigma*sqrt((1-rho)/pi).
# E(range_k) = 2*sigma*sqrt(1-rho)*k*integral z*phi(z)*Phi(z)^(k-1) dz.
# For k=5, E(max Zj) = 5/(4*sqrt(pi))*(1+6/pi*asin(1/3)).
#
# A second model demonstrates that even normal marginals and the entire
# correlation matrix do NOT determine these expectations without joint
# normality. With probability rho all scores share one normal draw;
# otherwise they are independent. Marginals/correlations are unchanged,
# but expected gaps/ranges scale by (1-rho), rather than sqrt(1-rho).
# This is a mathematical counterexample, not a proposed model of IQ tests.
#
# Numerical references (accessed 2026-09-03):
# https://stat.ethz.ch/R-manual/R-devel/library/stats/html/Normal.html
# https://stat.ethz.ch/R-manual/R-devel/library/stats/html/integrate.html
# https://stat.ethz.ch/R-manual/R-devel/library/stats/html/Tukey.html
# https://arxiv.org/abs/2004.04682 (normal orthant formula used in closed form)

options(width = 120)

parse_args <- function(args) {
  opt <- list(n = 5000000, rho = 0.6, sd = 15, mu = 100,
              k = 5, seed = 20260903, batch = 100000,
              out = "iq_simulation_results.txt")
  if ("--help" %in% args) {
    cat("Usage: Rscript iq_difference_simulation.R [--name=value ...]\n",
        "Options: --n=5000000 --rho=0.6 --sd=15 --mu=100 --k=5\n",
        "         --seed=20260903 --batch=100000 --out=iq_simulation_results.txt\n",
        "rho must be in [0,1]; the simulation uses a nonnegative common factor.\n",
        "The program refuses to overwrite an existing output file.\n", sep = "")
    quit(status = 0)
  }
  for (arg in args) {
    if (!grepl("^--[^=]+=.+$", arg)) stop("Use --name=value: ", arg)
    name <- sub("^--([^=]+)=.*$", "\\1", arg)
    value <- sub("^--[^=]+=", "", arg)
    if (!name %in% names(opt)) stop("Unknown option: ", name)
    opt[[name]] <- if (name == "out") value else suppressWarnings(as.numeric(value))
  }
  num <- unlist(opt[setdiff(names(opt), "out")])
  if (any(!is.finite(num))) stop("All numerical options must be finite.")
  for (key in c("n", "k", "batch", "seed")) {
    if (opt[[key]] != floor(opt[[key]])) stop(key, " must be an integer.")
  }
  if (opt$n < 2 || opt$n > 1e9) stop("n must be between 2 and 1e9.")
```

```
  if (opt$k < 2 || opt$k > 1000) stop("k must be between 2 and 1000.")
  if (opt$batch < 1) stop("batch must be positive.")
  if (opt$seed < 0 || opt$seed > .Machine$integer.max) stop("Invalid seed.")
  if (opt$rho < 0 || opt$rho > 1) stop("rho must be in [0,1].")
  if (opt$sd <= 0) stop("sd must be positive.")
  if (file.exists(opt$out)) stop("Output already exists; choose another --out: ", opt$out)
  if (!dir.exists(dirname(opt$out))) stop("The output directory does not exist.")
  opt
}

normal_max_mean <- function(k) {
  # Use infinite integration limits. Phi(z)^(k-1) is computed on the log scale.
  integrate(function(z) k * z * dnorm(z) *
              exp((k - 1) * pnorm(z, log.p = TRUE)),
            lower = -Inf, upper = Inf, rel.tol = 1e-11,
            subdivisions = 2000L, stop.on.error = TRUE)
}

empty_moments <- function(p, labels = NULL) {
  list(n = 0, mean = setNames(numeric(p), labels),
       M2 = setNames(numeric(p), labels))
}

merge_moments <- function(acc, x) {
  # Merge batch means/centered sums of squares, avoiding subtraction of
  # almost equal accumulated raw moments.
  nb <- nrow(x)
  mb <- colMeans(x)
  centered <- sweep(x, 2, mb, "-")
  m2b <- colSums(centered * centered)
  nt <- acc$n + nb
  delta <- mb - acc$mean
  acc$M2 <- acc$M2 + m2b + delta^2 * acc$n * nb / nt
  acc$mean <- acc$mean + delta * nb / nt
  acc$n <- nt
  acc
}

make_theory <- function(opt, max_mean, model) {
  shrink <- if (model == "gaussian") sqrt(1 - opt$rho) else 1 - opt$rho
  c(signed_difference = 0,
    absolute_difference = 2 * opt$sd * shrink / sqrt(pi),
    squared_difference = 2 * opt$sd^2 * (1 - opt$rho),
    test_range = 2 * opt$sd * shrink * max_mean,
    highest_score = opt$mu + opt$sd * shrink * max_mean,
    lowest_score = opt$mu - opt$sd * shrink * max_mean)
}

simulate_model <- function(opt, max_mean, model) {
  theory <- make_theory(opt, max_mean, model)
  acc <- empty_moments(length(theory), names(theory))
  score_n <- 0
  score_mean <- numeric(opt$k)
  score_M2 <- matrix(0, opt$k, opt$k)
  chunks <- ceiling(opt$n / opt$batch)
  for (b in seq_len(chunks)) {
    nb <- min(opt$batch, opt$n - acc$n)
    e <- matrix(rnorm(nb * opt$k), nrow = nb, ncol = opt$k)
    common <- rnorm(nb)
    if (model == "gaussian") {
      x <- opt$mu + opt$sd * (sqrt(opt$rho) * common + sqrt(1 - opt$rho) * e)
    } else {
      x <- opt$mu + opt$sd * e
      shared <- runif(nb) < opt$rho
      if (any(shared)) {
        shared_score <- opt$mu + opt$sd * common[shared]
```

```r
        x[shared, ] <- matrix(shared_score, nrow = sum(shared), ncol = opt$k)
      }
    }
    lo <- x[, 1]
    hi <- lo
    for (j in 2:opt$k) {
      lo <- pmin(lo, x[, j])
      hi <- pmax(hi, x[, j])
    }
    difference <- x[, 1] - x[, 2]
    statistics <- cbind(difference, abs(difference), difference^2, hi - lo, hi, lo)
    colnames(statistics) <- names(theory)
    acc <- merge_moments(acc, statistics)

    mb <- colMeans(x)
    centered <- sweep(x, 2, mb, "-")
    delta <- mb - score_mean
    nt <- score_n + nb
    score_M2 <- score_M2 + crossprod(centered) + tcrossprod(delta) * score_n * nb / nt
    score_mean <- score_mean + delta * nb / nt
    score_n <- nt
    if (b == 1 || b %% 10 == 0 || b == chunks) {
      message(sprintf("%s: %s / %s simulated people", model,
                      format(acc$n, big.mark = ",", scientific = FALSE),
                      format(opt$n, big.mark = ",", scientific = FALSE)))
    }
  }
  mc_sd <- sqrt(acc$M2 / (acc$n - 1))
  mc_se <- mc_sd / sqrt(acc$n)
  covariance <- score_M2 / (score_n - 1)
  table <- data.frame(
    quantity = names(theory), theory = unname(theory),
    simulation = unname(acc$mean), mc_se = unname(mc_se),
    mc_95_low = unname(acc$mean - qnorm(0.975) * mc_se),
    mc_95_high = unname(acc$mean + qnorm(0.975) * mc_se),
    error_in_mc_se = unname(ifelse(mc_se > 0, (acc$mean - theory) / mc_se, NA_real_))
  )
  list(table = table, means = score_mean, sds = sqrt(diag(covariance)),
       correlation = cov2cor(covariance), moments = acc)
}

opt <- parse_args(commandArgs(trailingOnly = TRUE))
RNGkind(kind = "Mersenne-Twister", normal.kind = "Inversion", sample.kind = "Rejection")
set.seed(opt$seed)

# Deterministic checks: closed forms, numerical quadrature, and covariance.
max_integral <- normal_max_mean(opt$k)
max_mean <- max_integral$value
max5_exact <- 5 / (4 * sqrt(pi)) * (1 + 6 / pi * asin(1 / 3))
max5_integral <- normal_max_mean(5)$value
max2_integral <- normal_max_mean(2)$value
stopifnot(abs(max5_integral - max5_exact) < 1e-9,
          abs(max2_integral - 1 / sqrt(pi)) < 1e-9,
          abs(2 * 15 * sqrt(0.4 / pi) - 10.7047446969166) < 1e-9)
target_cov <- opt$sd^2 * ((1 - opt$rho) * diag(opt$k) +
                         opt$rho * matrix(1, opt$k, opt$k))
factor_loadings <- opt$sd * cbind(rep(sqrt(opt$rho), opt$k),
                                sqrt(1 - opt$rho) * diag(opt$k))
stopifnot(max(abs(tcrossprod(factor_loadings) - target_cov)) < 1e-10 * opt$sd^2)

gaussian <- simulate_model(opt, max_mean, "gaussian")
mixture <- simulate_model(opt, max_mean, "mixture")

# Individual/model quantiles are different from Monte Carlo uncertainty.
probs <- c(0.025, 0.5, 0.975)
residual_sd <- opt$sd * sqrt(1 - opt$rho)
```

```
individual_quantiles <- data.frame(
  probability = probs,
  absolute_difference = sqrt(2) * residual_sd * qnorm((1 + probs) / 2),
  test_range = residual_sd * qtukey(probs, nmeans = opt$k, df = Inf)
)

check_message <- function(result) {
  # A 95% Monte Carlo interval need not cover its target on every run.
  # This is a diagnostic, not a rule to keep simulating until it passes.
  z <- result$table$error_in_mc_se
  if (any(abs(z[is.finite(z)]) > 6)) {
    "WARNING: at least one mean differs from theory by more than 6 MC SE; investigate."
  } else {
    "Monte Carlo check: every nondegenerate mean is within 6 MC SE of theory."
  }
}

report <- capture.output({
  cat("IQ TEST DIFFERENCES AND RANGES\n",
      "==============================\n\n", sep = "")
  cat(sprintf("People per model: %.0f; tests per person: %d; seed: %d; batch: %.0f\n",
              opt$n, opt$k, opt$seed, opt$batch))
  cat(sprintf("Marginal mean: %g; marginal SD: %g; every pair correlation: %g\n\n",
              opt$mu, opt$sd, opt$rho))
  cat("MODEL ASSUMPTIONS\n",
      "The main results assume an equicorrelated JOINT normal distribution.\n",
      "Each row is one person; independent rows define the averaging population.\n",
      "Scores are continuous and unrounded. Correlation is a population parameter,\n",
      "not the exact empirical correlation in the simulated finite sample.\n\n", sep = "")
  cat("DETERMINISTIC NUMERICAL CHECKS\n")
  cat(sprintf("E(max of %d independent standard normals): %.12f\n", opt$k, max_mean))
  cat(sprintf("Quadrature estimated absolute error: %.3g\n", max_integral$abs.error))
  cat(sprintf("E(max of 5), exact closed form: %.12f\n", max5_exact))
  cat(sprintf("Closed form minus quadrature: %.3g\n", max5_exact - max5_integral))
  cat("All deterministic checks passed.\n\n")
  cat("JOINT NORMAL MODEL\n")
  print(gaussian$table, row.names = FALSE, digits = 9)
  cat(check_message(gaussian), "\n\n")
  cat("Empirical test means:\n"); print(gaussian$means, digits = 8)
  cat("Empirical test SDs:\n"); print(gaussian$sds, digits = 8)
  cat("Empirical correlation matrix:\n"); print(gaussian$correlation, digits = 6)
  cat("\nINDIVIDUAL QUANTILES UNDER THE JOINT NORMAL MODEL\n")
  print(individual_quantiles, row.names = FALSE, digits = 8)
  cat("These are distribution quantiles, NOT confidence intervals for the mean.\n\n")
  cat("COUNTEREXAMPLE: NORMAL MARGINALS, SAME CORRELATIONS, NON-NORMAL JOINT LAW\n",
      "With probability rho all k scores equal one shared normal draw.\n",
      "Otherwise all k scores are independent normal draws.\n",
      "This is a non-identifiability demonstration, not a realistic test model.\n", sep = "")
  print(mixture$table, row.names = FALSE, digits = 9)
  cat(check_message(mixture), "\n\n")
  cat("Empirical counterexample test means:\n"); print(mixture$means, digits = 8)
  cat("Empirical counterexample test SDs:\n"); print(mixture$sds, digits = 8)
  cat("Empirical counterexample correlation matrix:\n"); print(mixture$correlation, digits = 6)
  cat("\nINTERPRETATION\n",
      "mc_se and mc_95_* quantify simulation error, not uncertainty in rho or\n",
      "the assumptions, and not the spread of an individual person's scores.\n",
      "squared_difference has units of IQ points squared; other means use IQ points.\n",
      "The 10 pairwise gaps within a row of 5 tests are not independent replicates.\n",
      "We use one specified pair and one range per independent person.\n",
      "The simulation supports implementation/numerics; the mathematical derivation\n",
      "proves the expectations under the stated model.\n\n", sep = "")
  cat("REPRODUCIBILITY\n"); print(sessionInfo())
})
writeLines(report, opt$out, useBytes = TRUE)
cat(paste(report, collapse = "\n"), "\n", sep = "")
message("Saved: ", normalizePath(opt$out, mustWork = TRUE))
```

C Complete extension and export source

C.1 R calculations, smooth-mixture simulation, and figures

```r
#!/usr/bin/env Rscript
# Supplement to iq_difference_simulation.R; base R only.
# Run from the report directory: Rscript code/iq_report_extensions.R
# Outputs: tables/*.csv and *.tex; figures/*.pdf; results/extensions.txt.
# These are model calculations, not empirical IQ-test observations.
options(width = 105)
for (d in c("tables", "figures", "results")) dir.create(d, showWarnings = FALSE)
sigma <- 15
rho <- 0.6
mu <- 100
k <- 5
s <- sigma * sqrt(1 - rho)
tau <- sqrt(2) * s
a <- function(k) integrate(function(z) k * z * dnorm(z) *
  exp((k - 1) * pnorm(z, log.p = TRUE)), -Inf, Inf,
  rel.tol = 1e-11, subdivisions = 2000)$value
a5 <- 5 / (4 * sqrt(pi)) * (1 + 6 / pi * asin(1 / 3))
# Finch (2016), section 1: second raw moment of the iid normal range.
nu5 <- 2 * (1 + 5 * sqrt(3) / (2 * pi) + 15 / pi^2 * acos(2 / 3) -
  5 * sqrt(3) / (2 * pi^2) * acos(1 / 4))
range_population_sd <- s * sqrt(nu5 - (2 * a5)^2)
folded_mean <- function(delta, tau) {
  2 * tau * dnorm(delta / tau) + delta * (2 * pnorm(delta / tau) - 1)
}
# Stable probability of a standard normal lying in [z,z+w].
normal_interval <- function(z, w) {
  pmax(0, ifelse(z > 0,
    pnorm(z, lower.tail = FALSE) - pnorm(z + w, lower.tail = FALSE),
    pnorm(z + w) - pnorm(z)))
}
range_cdf <- function(w, k = 5) {
  if (w <= 0) return(0)
  integrate(function(z) k * dnorm(z) * normal_interval(z, w)^(k - 1),
    -Inf, Inf, rel.tol = 1e-10, subdivisions = 2000)$value
}
save_table <- function(x, name, digits = 3) {
  write.csv(x, paste0("tables/", name, ".csv"), row.names = FALSE)
  rows <- vapply(seq_len(nrow(x)), function(i) {
    vals <- vapply(x, function(v) if (is.numeric(v))
      formatC(v[i], format = "f", digits = digits) else as.character(v[i]), "")
    paste0(paste(vals, collapse = " & "), " \\\\")
  }, "")
  writeLines(rows, paste0("tables/", name, ".tex"))
}
threshold <- c(5, 10, 15, 20, 30, 40)
probabilities <- data.frame(threshold = threshold,
  abs_exceeds_pct = 200 * pnorm(threshold / tau, lower.tail = FALSE),
  range_exceeds_pct = 100 * ptukey(threshold / s, k, Inf, lower.tail = FALSE))
save_table(probabilities, "probabilities", 2)
probs <- c(.025, .5, .95, .975)
quantiles <- data.frame(probability = probs,
  absolute_difference = tau * qnorm((1 + probs) / 2),
  range = s * qtukey(probs, k, Inf))
save_table(quantiles, "quantiles", 3)
rhos <- c(0, .3, .5, .6, .7, .8, .9, 1)
save_table(data.frame(rho = rhos,
  expected_abs = 2 * sigma * sqrt((1 - rhos) / pi),
  expected_range = 2 * sigma * sqrt(1 - rhos) * a5), "sensitivity", 3)
ks <- c(2, 3, 4, 5, 10, 20, 50)
```

```
save_table(data.frame(k = ks, expected_normal_max = vapply(ks, a, 0.0),
  expected_range = 2 * s * vapply(ks, a, 0.0)), "test_counts", 3)
xs <- c(85, 100, 115, 130, 145, 160)
delta <- (1 - rho) * (xs - mu)
save_table(data.frame(first_score = xs, expected_second = mu + rho * (xs - mu),
  expected_signed_gap = delta,
  expected_abs_gap = folded_mean(delta, sigma * sqrt(1 - rho^2))),
  "conditional", 3)
shrinks <- c(sqrt(.4), .4, mean(sqrt(1 - c(.3, .9))))
models <- data.frame(model = c("Joint Gaussian", "Shared/independent mixture",
  "Smooth Gaussian mixture"), expected_abs = 2 * sigma * shrinks / sqrt(pi),
  expected_range = 2 * sigma * shrinks * a5)
save_table(models, "models", 4)
# Independent numerical path for the normal range distribution and quantiles.
cdf_checks <- data.frame(w = c(.5, 1, 2, 3, 4, 6))
cdf_checks$integral <- vapply(cdf_checks$w, range_cdf, 0.0)
cdf_checks$ptukey <- ptukey(cdf_checks$w, 5, Inf)
cdf_checks$error <- cdf_checks$integral - cdf_checks$ptukey
q_integral <- vapply(probs, function(p)
  uniroot(function(w) range_cdf(w) - p, c(.001, 12), tol = 1e-9)$root, 0.0)
stopifnot(max(abs(cdf_checks$error)) < 1e-7,
  max(abs(q_integral - qtukey(probs, 5, Inf))) < 2e-4,
  abs(a(5) - a5) < 1e-10,
  abs(range_cdf(2, 2) - (2 * pnorm(sqrt(2)) - 1)) < 1e-9)

# A separate seeded experiment with continuous, positive-density components.
# Equal probabilities of equicorrelation .3 and .9 give marginal rho=.6.
RNGkind("Mersenne-Twister", "Inversion", "Rejection")
set.seed(20260904)
n <- 1000000
batch <- 100000
acc_n <- 0
acc_mean <- c(abs = 0, range = 0)
acc_M2 <- c(abs = 0, range = 0)
for (b in seq_len(n / batch)) {
  r <- ifelse(runif(batch) < .5, .3, .9)
  common <- rnorm(batch)
  e <- matrix(rnorm(batch * 5), batch, 5)
  x <- sigma * (sqrt(r) * common + sqrt(1 - r) * e)
  lo <- hi <- x[, 1]
  for (j in 2:5) { lo <- pmin(lo, x[, j]); hi <- pmax(hi, x[, j]) }
  y <- cbind(abs = abs(x[, 1] - x[, 2]), range = hi - lo)
  mb <- colMeans(y)
  m2b <- colSums(sweep(y, 2, mb)^2)
  nt <- acc_n + batch
  change <- mb - acc_mean
  acc_M2 <- acc_M2 + m2b + change^2 * acc_n * batch / nt
  acc_mean <- acc_mean + change * batch / nt
  acc_n <- nt
}
smooth_check <- data.frame(quantity = names(acc_mean),
  theory = unlist(models[3, 2:3], use.names = FALSE),
  simulated = unname(acc_mean), mc_se = sqrt(acc_M2 / (n - 1) / n))
smooth_check$z_mc <- (smooth_check$simulated - smooth_check$theory) / smooth_check$mc_se
save_table(smooth_check, "smooth_simulation", 5)

# Full admissible equicorrelation range: verify projection construction.
H <- diag(5) - matrix(1 / 5, 5, 5)
for (r in c(-.25, -.1, 0, .6, 1)) {
  L <- cbind(sqrt(1 - r) * H, rep(sqrt((1 + 4 * r) / 5), 5))
  target <- (1 - r) * diag(5) + r * matrix(1, 5, 5)
  stopifnot(max(abs(tcrossprod(L) - target)) < 1e-12)
}
# Average correlation .6, but four pairs at 1 and six pairs at 1/3.
average_rho <- (4 + 6 / 3) / 10
block_range <- 2 * sigma * sqrt((1 - 1 / 3) / pi)
```

```
# Reproducible vector figures with the exact plotted values in CSV.
ink <- "#183B56"
teal <- "#007F82"
rust <- "#B75D36"
rho_grid <- seq(0, 1, length.out = 201)
rho_curve <- data.frame(rho = rho_grid,
  absolute = 2 * sigma * sqrt((1 - rho_grid) / pi),
  range = 2 * sigma * sqrt(1 - rho_grid) * a5)
write.csv(rho_curve, "tables/figure_correlation.csv", row.names = FALSE)
cairo_pdf("figures/correlation.pdf", width = 7.1, height = 3.7, family = "DejaVu Sans")
par(mar = c(4.2, 4.4, 1.1, .7), las = 1)
matplot(rho_curve$rho, rho_curve[, 2:3], type = "l", lty = 1, lwd = 2.2,
  col = c(teal, ink), xlab = "Common pairwise correlation",
  ylab = "Expected gap (IQ points)", ylim = c(0, 36))
abline(v = .6, col = "#AAAAAA", lty = 3)
points(rep(.6, 2), c(2 * s / sqrt(pi), 2 * s * a5), pch = 19, col = c(teal, ink))
legend("topright", c("Five-test range", "Two-test absolute difference"),
  col = c(ink, teal), lwd = 2.2, bty = "n", cex = .88)
dev.off()
q <- seq(0, 60, by = .15)
distributions <- data.frame(gap = q, abs_cdf = 2 * pnorm(q / tau) - 1,
  range_cdf = ptukey(q / s, 5, Inf))
write.csv(distributions, "tables/figure_distributions.csv", row.names = FALSE)
cairo_pdf("figures/distributions.pdf", width = 7.1, height = 3.7, family = "DejaVu Sans")
par(mar = c(4.2, 4.4, 1.1, .7), las = 1)
matplot(q, distributions[, 2:3], type = "l", lty = 1, lwd = 2.2,
  col = c(teal, ink), xlab = "Gap (IQ points)",
  ylab = "Probability gap is at most this value", ylim = c(0, 1))
abline(h = c(.025, .5, .975), col = "#DDDDDD", lty = 3)
legend("bottomright", c("Two-test absolute difference", "Five-test range"),
  col = c(teal, ink), lwd = 2.2, bty = "n", cex = .88)
dev.off()
xg <- seq(60, 160, by = .5)
d <- (1 - rho) * (xg - mu)
conditional_curve <- data.frame(first_score = xg,
  signed_gap = d, absolute_gap = folded_mean(d, sigma * sqrt(1 - rho^2)))
write.csv(conditional_curve, "tables/figure_conditional.csv", row.names = FALSE)
cairo_pdf("figures/conditional.pdf", width = 7.1, height = 3.7, family = "DejaVu Sans")
par(mar = c(4.2, 4.4, 1.1, .7), las = 1)
matplot(xg, conditional_curve[, 2:3], type = "l", lty = c(2, 1), lwd = 2.2,
  col = c(rust, ink), xlab = "Observed first-test score",
  ylab = "Conditional expected gap (IQ points)")
abline(h = 0, col = "#DDDDDD")
legend("topleft", c("Absolute gap", "Signed gap: first minus second"),
  col = c(ink, rust), lty = c(1, 2), lwd = 2.2, bty = "n", cex = .85)
dev.off()
log <- capture.output({
  cat("REPORT EXTENSIONS: 2026-09-03\nNo empirical IQ data used.\n\n")
  cat("Range CDF quadrature checks:\n"); print(cdf_checks, digits = 12)
  cat("Range quantiles from independent CDF inversion (IQ points):\n")
  print(data.frame(probability = probs, quantile = s * q_integral), digits = 12)
  cat("\nSmooth mixture: n=1000000, seed=20260904\n")
  print(smooth_check, digits = 10, row.names = FALSE)
  cat("\nAverage-correlation block model: average rho =", average_rho,
    "; expected range =", block_range, "\n")
  cat("\nProjection covariance checks passed at rho=-.25,-.1,0,.6,1.\n")
  cat("Exact iid normal range second moment =", format(nu5, digits = 12), "\n")
  cat("Gaussian five-test range population SD =",
    format(range_population_sd, digits = 12), "\n")
  cat("Theoretical range MC SE at n=5000000 =",
    format(range_population_sd / sqrt(5000000), digits = 12), "\n")
  cat("\nAll deterministic checks passed.\n\n"); print(sessionInfo())
})
writeLines(log, "results/extensions.txt")
cat(paste(log, collapse = "\n"), "\n")
```

C.2 Companion workbook export

```
// Optional companion workbook exporter; does not run simulations.
// Requires @oai/artifact-tool. Run with Node and the report directory argument.
import fs from 'node:fs/promises';
import path from 'node:path';
import { Workbook, SpreadsheetFile } from '@oai/artifact-tool';
process.on('uncaughtException', error => {
  console.error(error.message);
  console.error(error.stack.split('\n').slice(-5).join('\n'));
  process.exit(1);
});

const root = path.resolve(process.argv[2] || '.');
const outputDir = path.join(root, 'outputs', '7beec1facbc5');
const qaDir = path.join(root, 'qa');
await fs.mkdir(outputDir, { recursive: true });
await fs.mkdir(qaDir, { recursive: true });
const wb = Workbook.create();
const inputs = wb.worksheets.add('Model');
inputs.showGridLines = false;
inputs.getRange('A1:C15').values = [
  ['Quantity', 'Value', 'Meaning / units'],
  ['Population mean', 100, 'IQ points; assumed'],
  ['Common SD', 15, 'IQ points; assumed'],
  ['Every pair correlation', 0.6, 'Population parameter; assumed'],
  ['Number of tests', 5, 'Five-test closed form below'],
  ['Expected maximum of 5 iid N(0,1)', null, 'Exact inverse-trigonometric formula'],
  ['Expected absolute pair gap', null, 'IQ points; jointly normal model'],
  ['Expected five-test range', null, 'IQ points; jointly normal model'],
  ['Difference SD / RMS', null, 'IQ points; equal means'],
  ['Source', 'Report: absolute gap and five-test range formulas', 'See accompanying PDF and R code'],
  ['Data status', 'Synthetic / theoretical', 'No measured IQ scores in this workbook'],
  ['Other sheets', 'Fixed R exports', 'They preserve the report run and do not update with inputs'],
  ['Recompute exports', 'Rscript code/iq_report_extensions.R', 'Then rerun this exporter'],
  ['Range source', 'https://arxiv.org/pdf/1111.4976', 'Finch, section 1'],
  ['CDF source', 'https://stat.ethz.ch/R-manual/R-devel/library/stats/html/Tukey.html', 'R ptukey / qtukey; df=
    Inf']
];
inputs.getRange('B6:B9').formulas = [
  ['=5/(4*SQRT(PI()))*(1+6/PI()*ASIN(1/3))'],
  ['=2*B3*SQRT((1-B4)/PI())'],
  ['=IF(B5=5,2*B3*SQRT(1-B4)*B6,"Requires k=5")'],
  ['=B3*SQRT(2*(1-B4))']
];
inputs.getRange('A1:C15').format.font.name = 'Arial';
inputs.getRange('A1:C15').format.font.size = 11;
inputs.getRange('A1:C15').format.wrapText = true;
inputs.getRange('A1:A15').format.columnWidth = 38;
inputs.getRange('B1:B15').format.columnWidth = 48;
inputs.getRange('C1:C15').format.columnWidth = 52;
inputs.getRange('A1:C15').format.rowHeight = 38;
inputs.getRange('B2:B9').setNumberFormat('0.0000');
inputs.getRange('B5').setNumberFormat('0');
inputs.getRange('C15').clear({ applyTo: 'contents' });
inputs.getRange('B15:C15').merge();
inputs.getRange('B2:B5').format.font.color = '#176B88';
inputs.getRange('B6:B9').format.fill = '#E7F3F2';
inputs.getRange('A1:C1').format = {
  fill: '#183B56', font: { bold: true, color: '#FFFFFF' }, rowHeight: 30
};
inputs.freezePanes.freezeRows(1);

const exports = [
  ['probabilities', 'Threshold probabilities', 'Percent exceeding threshold; IQ-point thresholds'],
  ['quantiles', 'Quantiles', 'Probability; IQ-point quantiles'],
```

```
  ['sensitivity', 'Correlation sensitivity', 'Correlation; expected IQ-point gaps'],
  ['test_counts', 'Number of tests', 'Test count; standard-normal maximum; IQ-point range'],
  ['conditional', 'Conditional scores', 'All score and gap values in IQ points'],
  ['models', 'Alternative models', 'Expected gaps in IQ points'],
  ['smooth_simulation', 'Smooth mixture run', 'IQ-point means and MC SE; dimensionless z_mc'],
  ['figure_correlation', 'Correlation curve', 'Exact values plotted in correlation.pdf'],
  ['figure_distributions', 'Distribution curves', 'IQ-point gaps and CDF probabilities'],
  ['figure_conditional', 'Conditional curves', 'IQ-point first scores and expected gaps']
];
for (const [file, sheetName, units] of exports) {
  const csv = await fs.readFile(path.join(root, 'tables', file + '.csv'), 'utf8');
  // Import into a fresh workbook, then transfer typed values. Importing CSV
  // into an already populated collaborative workbook is not supported.
  const imported = await Workbook.fromCSV(csv, { sheetName });
  const sheet = wb.worksheets.add(sheetName);
  sheet.getRange('A1').writeValues(imported.worksheets.getItemAt(0).getUsedRange().values);
  sheet.showGridLines = false;
  const used = sheet.getUsedRange();
  used.format.font.name = 'Arial';
  used.format.font.size = 11;
  used.format.columnWidth = file === 'models' ? 35 : 27;
  used.format.rowHeight = 24;
  used.setNumberFormat('0.000000');
  if (file === 'test_counts' || file === 'probabilities') {
    sheet.getRange('A2:A' + csv.trim().split('\n').length).setNumberFormat('0');
  }
  const count = csv.trim().split('\n')[0].split(',').length;
  sheet.getRangeByIndexes(0, 0, 1, count).format = {
    fill: '#183B56', font: { bold: true, color: '#FFFFFF' },
    wrapText: true, rowHeight: 48
  };
  const rowCount = csv.trim().split('\n').length;
  sheet.getCell(rowCount + 1, 0).values = [['Units / provenance']];
  sheet.getCell(rowCount + 2, 0).values = [[units]];
  sheet.getRangeByIndexes(rowCount + 2, 0, 1, count).merge();
  sheet.getRangeByIndexes(rowCount + 2, 0, 1, count).format.wrapText = true;
  sheet.getRangeByIndexes(rowCount + 2, 0, 1, count).format.rowHeight = 34;
  sheet.getCell(rowCount + 3, 0).values = [['Source: code/iq_report_extensions.R; model calculations, 2026-09-03
    ']];
  sheet.getRangeByIndexes(rowCount + 3, 0, 1, count).merge();
  sheet.getRangeByIndexes(rowCount + 3, 0, 1, count).format.wrapText = true;
  sheet.getRangeByIndexes(rowCount + 3, 0, 1, count).format.rowHeight = 34;
  sheet.freezePanes.freezeRows(1);
}
console.log((await wb.inspect({ kind: 'table', range: 'Model!A1:C9',
  include: 'values,formulas', tableMaxRows: 9, tableMaxCols: 3,
  maxChars: 3000 })).ndjson);
console.log((await wb.inspect({ kind: 'match',
  searchTerm: '#REF!|#DIV/0!|#VALUE!|#NAME\\?|#N/A',
  options: { useRegex: true, maxResults: 30 }, maxChars: 1000,
  summary: 'Formula error scan' })).ndjson);
for (const name of ['Model', ...exports.map(x => x[1])]) {
  const sheet = wb.worksheets.getItem(name);
  const last = name === 'Model' ? 'C15' : name === 'Smooth mixture run' ? 'E8' : 'D8';
  const blob = await wb.render({ sheetName: name, range: 'A1:' + last, scale: 1.5 });
  await fs.writeFile(path.join(qaDir, 'workbook_' + name.replaceAll(' ', '_') + '.png'),
    new Uint8Array(await blob.arrayBuffer()));
}
await (await SpreadsheetFile.exportXlsx(wb)).save(path.join(outputDir, 'iq_report_data.xlsx'));
console.log('Saved iq_report_data.xlsx');
```

D Complete AI-use disclosure

Authorship and review status

This report was generated with ChatGPT / Codex, provided by OpenAI, at the request of Anders Hellström. AI assistance was substantive throughout the research, mathematical presentation, programming, analysis, visualization, and document production. It was not limited to proofreading or formatting. No independent human mathematical or psychometric peer review is claimed.

D.1 Human contribution documented in the conversation

The user posed the original questions about a two-test difference and a five-test range at correlation 0.6, requested rigorous proof and an R simulation, and subsequently requested an expanded, thoroughly sourced report. The user specified XeLaTeX, KOMA-Script `scrbook`, narrow margins, the principal packages, an unmissable opening AI disclosure, a complete closing disclosure, and source-code appendices. The user did not supply an observed IQ-score dataset or a set of named tests for empirical validation. The numerical SD-15 and equal-mean conventions were modelling assumptions made explicit in the preceding AI-assisted analysis and retained here.

D.2 AI contribution

The coordinating AI assistant recovered and read the preceding proof and R script, assessed the assumptions, researched relevant public sources, and developed the report's narrative and mathematical exposition. It verified the two-test calculation, the cancellation argument for ranges, and the five-normal closed form. It added explanations of the complete range distribution, conditional prediction, sensitivity, bounds, a smooth-mixture counterexample, an average-correlation counterexample, and the extension to negative equicorrelation.

Two additional AI research workers performed bounded searches on mathematical sources and psychometric interpretation. Their source records were reviewed and integrated by the coordinating assistant. The workers used the same general AI workflow and are not independent human validators. The exact underlying model build identifier was not independently recorded for the report; no unverified model-version label is asserted.

The AI wrote the supplementary R and workbook-export code, reran the original R program, executed the supplementary computations, generated the vector figures and numerical tables, and authored the XeLaTeX document and bibliography. All newly authored computational programs are supplied in full. The preceding main R script itself was also AI-generated, as disclosed in the earlier conversation, and is preserved unchanged in this report package.

D.3 Data, outputs, and checks actually performed

No real participants were enrolled, and no observed IQ-test scores were analysed. Every numeric data row used for simulation is pseudorandomly generated under one of the stated models. The displayed simulation estimates are from executed R code, with fixed seeds and saved session information. Literature sample sizes and study findings are separately cited and were not merged into the synthetic data.

The principal formulas were checked by analytic derivation, a published-source comparison, quadrature, and Monte Carlo. The range CDF was evaluated by an independent integral implementation as well as

R's studentized-range functions. The exact range second moment is explicitly attributed to Finch rather than presented as independently proved. The repeated main run reproduced the earlier log exactly. The supplementary smooth-mixture run used a different fixed seed. Sources available only as abstracts and a result read in a preliminary paper version are identified in the evidence chapter and references.

The PDF was compiled with XeLaTeX, checked for missing citations and layout warnings, and visually reviewed on selected high-risk pages. This was sampled visual inspection, not a claim that a human reviewed every page. Automated structural checks and sampled visual checks do not establish mathematical or empirical validity.

D.4 Limits and responsibility

AI systems can produce incorrect deductions, code, citations, or interpretations. The checks documented above reduce particular risks but cannot exclude all errors. Agreement among AI-generated derivations and simulations is especially limited when both may share the same assumptions. The externally sourced exact formula and the distinct numerical CDF implementation provide additional checks, while applicability to actual IQ tests remains unverified.

The report is a mathematical and methodological analysis, not a personal IQ assessment, a diagnosis, or evidence that particular real tests follow the stated model. It claims no original discovery of the classical normal-range or Gaussian orthant formulas. A reader intending to cite, publish, or apply the report should evaluate its assumptions and calculations and retain the substantive AI disclosure. Anders Hellström is identified as the requester; the report does not attribute unobserved human review, endorsement, or authorship of the AI-generated technical material to him.

End of report.